bash - python main.py --demo (Topic 07 Safe Agent Execution)

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      Topic 07 - Autonomous Agent & Tool Integration Demo
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>>> Scenario 1: Multi-Step Agent Loop (Customer finds in-stock laptop)
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User Request : Find a laptop that is currently in stock.
Active Role  : CUSTOMER
LLM Provider : SIMULATED
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[Step 1] Agent Reasoning...
  Thought : The user is searching for items. Searching catalog for 'laptop'.
  Decision: Propose Tool -> search_products({'query': 'laptop'})
  Harness : Checked -> Status: success
  Result  : {'status': 'success', 'count': 2, 'results': [{'id': 1, 'name': 'ThinkPad
Carbon Ultrabook', 'category': 'Laptops', 'price': 1299.99}, {'id': 2, 'name': 'MacBook
Air 13-inch', 'category': 'Laptops', 'price': 999.0}]}

[Step 2] Agent Reasoning...
  Thought : Catalog search returned results. Now checking inventory stock for candidate
ID 1.
  Decision: Propose Tool -> check_stock({'product_id': 1})
  Harness : Checked -> Status: success
  Result  : {'status': 'success', 'product_id': 1, 'name': 'ThinkPad Carbon Ultrabook',
'stock': 5, 'in_stock': True}

[Step 3] Agent Reasoning...
  Thought : We have identified the product and verified current stock availability.
  Decision: Final Answer
  Answer  : I found the 'ThinkPad Carbon Ultrabook'. It is currently in stock with 5
units available for purchase.


>>> Scenario 2: Permission Control Check (Customer attempts delete_product)
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User Request : Delete product 1 from the catalog.
Active Role  : CUSTOMER
LLM Provider : SIMULATED
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[Step 1] Agent Reasoning...
  Thought : The user wants to delete product ID 1. Proposing 'delete_product' action.
  Decision: Propose Tool -> delete_product({'product_id': 1})
  Harness : Checked -> Status: error
  Result  : {'status': 'error', 'error_code': 'PERMISSION_DENIED', 'message':
"Permission denied: role 'customer' is not allowed to call 'delete_product'."}
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[Step 2] Agent Reasoning...
  Thought : Action was rejected by application: Permission denied: role 'customer' is
not allowed to call 'delete_product'.
  Decision: Final Answer
  Answer  : Unable to delete product 1. Permission denied: role 'customer' is not
allowed to call 'delete_product'.


>>> Scenario 3: Authorized Execution (Admin deletes product 1)
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User Request : Delete product 1 from the catalog.
Active Role  : ADMIN
LLM Provider : SIMULATED
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[Step 1] Agent Reasoning...
  Thought : The user wants to delete product ID 1. Proposing 'delete_product' action.
  Decision: Propose Tool -> delete_product({'product_id': 1})
  Harness : Checked -> Status: success
  Result  : {'status': 'success', 'message': "Product 'ThinkPad Carbon Ultrabook' (ID:
1) was successfully deleted from catalog."}

[Step 2] Agent Reasoning...
  Thought : Observed successful deletion from catalog.
  Decision: Final Answer
  Answer  : Product 'ThinkPad Carbon Ultrabook' (ID: 1) was successfully deleted from
catalog.


>>> Scenario 4: Safety Validation (Negative ID or invalid quantity)
Executing tool with invalid argument product_id=-99...
Validation Result: {'status': 'error', 'error_code': 'VALIDATION_ERROR', 'message':
"Input validation failed for 'check_stock': product_id: Input should be greater than 0"}

>>> Scenario 5: Maximum Tool-Call Limit Boundary
Call 1 status: success
Call 2 (exceeding limit) status: {'status': 'error', 'error_code':
'RATE_LIMIT_EXCEEDED', 'message': 'Execution halted: reached maximum limit of 1 tool
calls.'}

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All demonstration scenarios completed successfully.
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